

CHAPTER -3

PATTERN PRINTING WITH LOOPS

- Pattern printing in loops is usually done with a **nested loop**: the outer loop controls rows, and the inner loop controls how many stars, numbers, or spaces are printed in each row. Here are a few simple Java examples with output.

1) Square star pattern:

```
public class Main {  
    public static void main(String[] args) {  
        int n = 4;  
  
        for (int i = 1; i <= n; i++) {  
            for (int j = 1; j <= n; j++) {  
                System.out.print("* ");  
            }  
            System.out.println();  
        }  
    }  
}
```

Output:

* * * *

* * * *

* * * *

* * * *

2) Right triangle star pattern:

```
public class Main {  
    public static void main(String[] args) {  
        int n = 5;  
  
        for (int i = 1; i <= n; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print("* ");  
            }  
            System.out.println();  
        }  
    }  
}
```

Output:

```
*  
  
* *  
  
* * *  
  
* * * *  
  
* * * * *
```

3) Inverted triangle star pattern:

```
public class Main {  
    public static void main(String[] args) {  
        int n = 5;  
  
        for (int i = n; i >= 1; i--) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print("* ");  
            }  
            System.out.println();  
        }  
    }  
}
```

Output:

* * * * *

* * * *

* * *

* *

*

4) Number triangle pattern:

```
public class Main {  
    public static void main(String[] args) {  
        int n = 5;  
  
        for (int i = 1; i <= n; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print(j + " ");  
            }  
            System.out.println();  
        }  
    }  
}
```

Output:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

5) Pyramid pattern:

```
public class Main {
    public static void main(String[] args) {
        int n = 5;

        for (int i = 1; i <= n; i++) {
            for (int s = 1; s <= n - i; s++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= (2 * i - 1); j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

```
    }  
  }  
}
```

Output:

```
  *  
  
 ***  
  
*****  
  
*****  
  
*****
```